Catalogue of Catalogues: A Web Platform for Earthquake Catalogue Management, Merging, and Statistical Analysis

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Abstract

We present the *Catalogue of Catalogues* (CofC), an open-source web platform for ingesting, quality-assessing, merging, and statistically analysing earthquake catalogues from multiple agencies and formats, designed in accordance with the FAIR data principles (Findable, Accessible, Interoperable, Reusable). CofC accepts data as CSV/TXT, GeoJSON, QuakeML 1.2 Basic Event Description (BED), and live FDSN web-service streams. A five-strategy merging engine detects duplicate events using configurable, adaptive time ($\leq 60\,\mathrm{s}$) and distance ($\leq 50\,\mathrm{km}$) windows (widened for larger and deeper events), with magnitude consistency enforced as a separate per-group gate, and resolves conflicts via priority, quality-score, newest-data, most-complete, or averaged-value rules. A transparent quality-scoring system (0–100 index, A^+ to F letter grade) evaluates each event across five dimensions: location precision, network geometry, solution fit, magnitude quality, and analyst-evaluation status. Standard seismological analyses, Gutenberg-Richter b-value estimation via maximum likelihood, magnitude of completeness (M_c) via the Maximum Curvature (MAXC) method, Gardner-Knopoff and Reasenberg window declustering, and cumulative seismic moment, are available in the browser. Using a fully synthetic two-catalogue New Zealand example (no real event data), we illustrate the end-to-end workflow and verify that it is internally consistent and reproducible: an injected 50% duplicate overlap is reproduced, quality scoring isolates an offshore high-azimuthal-gap population, and Gutenberg-Richter analysis of the quality-filtered catalogue returns $\hat{b} = 1.02$ (formal ± 0.005; realistic uncertainty larger) with full provenance retained. These values are properties of the controlled synthetic inputs, not inferences about real New Zealand seismicity. Six categories of statistical bias affecting merged-catalogue analysis are identified and actionable mitigation guidance is provided. Source code and documentation are freely available under the MIT licence.

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1 Introduction

1.1 The role of earthquake catalogues in seismology

Earthquake catalogues are the fundamental observational basis for quantitative seismology. Every tool in the modern seismological toolkit, from probabilistic seismic hazard analysis (PSHA; Cornell 1968; e.g. the 2022 Aotearoa New Zealand NSHM, Gerstenberger et al. 2024) to operational earthquake forecasting (Schorlemmer et al., 2007) to tectonic geodesy, rests on the assumption that the underlying catalogue faithfully represents the space-time-magnitude distribution of seismicity. The Gutenberg-Richter (GR) frequency-magnitude relation (Gutenberg and Richter, 1944) and its b-value, which encodes the relative proportion of small to large earthquakes (Kagan, 1999; Schorlemmer et al., 2005), are estimated directly from catalogue data; any systematic error in the catalogue propagates directly into hazard and risk assessments. Similarly, the Epidemic-Type Aftershock Sequence (ETAS) model (Ogata, 1988), the most widely used stochastic framework for short-term earthquake forecasting, requires a complete, Poissonian background catalogue that is free from both incompleteness and aftershock contamination.

Regional monitoring agencies, GeoNet (New Zealand; Petersen et al. 2011), the Northern California Earthquake Data Center (Northern California Earthquake Data Center, 2014), the International Seismological Centre (ISC; Storchak et al. 2013), the Engdahl-van der Hilst-Buland relocated global catalogue (Engdahl et al., 1998), each maintain catalogues with overlapping geographic coverage but different station networks, velocity models, magnitude scales, analysis procedures, and update frequencies. Researchers routinely need to *merge* or *compare* these sources: to extend a short modern catalogue with historical data, to supplement a global catalogue with a higher-resolution regional one, or to construct a unified input for PSHA over a region monitored by multiple agencies. This multi-source workflow introduces three classes of problem that, if ignored, produce statistically invalid results.

1.2 Problem 1: Format heterogeneity

No universally adopted exchange format exists for seismological data. The QuakeML 1.2 Basic Event Description (BED) schema (Schorlemmer et al., 2011) provides a comprehensive XML vocabulary, covering origins, magnitudes, focal mechanisms, picks, arrivals, and amplitude measurements, and is the standard output of the FDSN Event Web Service (International Federation of Digital Seismograph Networks, 2013). However, many agency archives remain in agency-specific CSV layouts, plain text bulletins, or legacy binary formats. The GeoJSON format (Butler et al., 2016) has gained traction for web-facing services but provides no controlled vocabulary for seismological fields; column and property names vary freely between providers. When a researcher assembles a multi-source dataset, the practical first step is writing bespoke format-conversion code, a task that is repetitive, error-prone, and rarely shared or version-controlled.

1.3 Problem 2: Duplicate events and location scatter

When two catalogues cover the same region, the same earthquake typically appears in both, reported with slightly different origin times, hypocentres, and magnitudes. These differences arise from genuine physical causes: different station sets lead to different travel-time residuals; regional versus global velocity models bias depth systematically; automatic picks differ from manually reviewed ones. The global relocation literature documents that, even for well-recorded teleseismic events, agency-to-agency hypocentre differences commonly reach 10–20 km in epicentre and tens of kilometres in the poorly constrained depth (Engdahl et al., 1998). Naïvely concatenating catalogues without duplicate removal inflates $N(M)$ at all magnitudes above the lower agency's M_c, lowers the apparent a-value, and distorts rate estimates. Storchak et al. (2013) describe the careful duplicate-association procedure developed for the ISC-GEM global catalogue, which required multi-year expert review. Automated, configurable duplicate association is available in catalogue-building libraries (GEM Foundation, 2024) and operational monitoring systems (GFZ German Research Centre for Geosciences and gempa GmbH, 2008), but not in a browser-based interface that a regional researcher can apply directly to an arbitrary pair of uploaded catalogues without scripting or an operational deployment.

1.4 Problem 3: Statistical biases in merged catalogues

Even after format conversion and duplicate removal, several statistical biases require active mitigation before standard analyses are valid. We summarise the mechanisms here; the corresponding practical mitigation steps, and how CofC supports each, are collected in Section 7 and not repeated in between.

Magnitude of completeness (M_c). A catalogue is complete above M_c, the smallest magnitude at which essentially all events are detected by the network. Below M_c, the GR relation breaks down (Rydelek and Sacks, 1989). Fitting b without first determining and enforcing M_c leads to systematic underestimation (Woessner and Wiemer, 2005). Multiple methods for estimating M_c have been proposed (Wiemer and Wyss, 2000; Cao and Gao, 2002; Woessner and Wiemer, 2005; Mignan and Woessner, 2012), each with different assumptions and performance characteristics. When merging two catalogues with different M_c values (e.g., $M_c = 1.5$ for a dense regional network and $M_c = 3.0$ for a global catalogue), the combined FMD will be incomplete between the two thresholds unless the analyst enforces the higher cut-off.

Magnitude scale heterogeneity. Different agencies routinely use different magnitude scales: local magnitude (M_L; Richter 1935), body-wave magnitude (m_b), surface-wave magnitude (M_s), and moment magnitude (M_w; Hanks and Kanamori 1979; Kanamori 1977). Each scale saturates at a different value, M_L above ~ 6.5, m_b above ~ 6.0 (Bormann and Saul, 2009), and the inter-scale regressions carry substantial scatter that depends on the regression method adopted (Das et al., 2011; Castellaro et al., 2006). Mixing scales inflates the apparent variance of the FMD, artificially shifts M_c, and can change the estimated b by up to ± 0.2 (Marzocchi and Sandri, 2003). Tinti and Mulargia (1987) showed that even with a single scale, binning width and sample size both affect the precision of $\hat{b}$ in ways that are routinely underappreciated.

Aftershock clustering. Aftershock sequences violate the Poissonian independence assumption underpinning the GR relation and ETAS background estimation. Aftershocks are most numerous at the small magnitudes just above M_c, where they inflate event counts; because clustered aftershock populations typically carry a higher b-value than the background, failing to decluster biases the apparent $\hat{b}$ of the raw catalogue *upward* relative to the declustered (Poissonian) catalogue, although the sign and size of the shift depend on the relative size distributions of the clustered and background populations (Gardner and Knopoff, 1974; Marsan and Lengliné, 2008). We illustrate this shift directly in Section 6. Multiple declustering algorithms exist, the window method of Gardner and Knopoff (1974) and the aftershock-zone scaling of Kagan (2002), the link-based method of Reasenberg (1985), and the ETAS-based stochastic method of Zhuang et al. (2002), with different sensitivities to swarm-type seismicity and to the aftershocks of aftershocks (van Stiphout et al., 2012). Helmstetter et al. (2005) demonstrated that ignoring small-magnitude aftershocks, often the majority of any sequence, underestimates cascading stress transfer and therefore the effective rate of background seismicity. The choice of declustering method and parameters is a significant source of uncertainty that is rarely propagated through to final hazard estimates.

Location quality filtering. Events recorded by sparse networks have large location uncertainties and poorly constrained depths. If such events are systematically concentrated at one magnitude or depth range, as is typical for offshore events recorded by onshore networks, they introduce spatial bias into b-value mapping and Gutenberg-Richter fits. Bondár et al. (2004) showed that network geometry, in particular the *secondary* azimuthal gap, together with the number of stations and the nearest-station distance, controls horizontal location accuracy: large gaps (e.g. a primary gap exceeding 180°) are typically associated with errors of tens of kilometres.

1.5 Existing tools and their limitations

The standard desktop tool for catalogue-based seismicity analysis is ZMAP (Wiemer, 2001), which implements GR analysis, spatial b-value mapping, declustering, and Omori-law fitting in a MATLAB environment. While scientifically comprehensive, ZMAP requires MATLAB licensing and familiarity with its GUI, and it does not support multi-catalogue merging or quality-scored event selection. ObsPy (Beyreuther et al., 2010) provides a Python framework for reading and writing QuakeML and performing basic catalogue operations, but all analysis requires custom scripting. EQcorrscan (Chamberlain et al., 2018) enables template-matching and repeating-earthquake detection but produces catalogues that then require separate quality assessment and merging tools. OpenQuake (Pagani et al., 2014) consumes pre-processed catalogues for PSHA but provides no tools for the preprocessing itself. Neither ZMAP, ObsPy, nor OpenQuake offers a guided workflow for avoiding the biases described above, nor a persistent web-accessible database for storing and sharing processed catalogues. The Community Online Resource for Statistical Seismicity Analysis (CORSSA; Naylor et al., 2010; Mignan and Woessner, 2012; van Stiphout et al., 2012; Woessner et al., 2010) provides peer-reviewed tutorials on these methods but no software platform.

The need for such integration is intensifying. Template-matching and machine-learning phase-detection workflows now routinely yield catalogues with up to an order of magnitude more events than traditional analyst-reviewed bulletins (Chamberlain et al., 2018; Mousavi and Beroza, 2022; Illsley-Kemp and Mestel, 2025; Michailos et al., 2025; Chamberlain et al., 2025), while every regional network continues to publish its own authoritative product. A single study area may therefore be covered by several catalogues that differ in completeness, magnitude scale, and location quality and must be reconciled before analysis. The constituent operations, format conversion, duplicate association, completeness estimation, declustering, are individually well understood and, in isolation, available in existing tools. The GEM OpenQuake Model Building Toolkit (GEM Foundation, 2024) compiles and homogenises multi-source catalogues with configurable space–time duplicate windows, declustering, and completeness estimation; SeisComP (GFZ German Research Centre for Geosciences and gempa GmbH, 2008) performs automated multi-agency origin association and merging within operational monitoring pipelines. What is missing is therefore not the algorithms but a single, *browser-native, no-install* platform that chains ingest, configurable merging, transparent per-event quality scoring, and bias-aware analysis behind one interface, preserves provenance at every step, and is usable by a regional researcher on an arbitrary pair of uploaded catalogues without scripting or a configured monitoring deployment. Table 1 contrasts the capabilities of representative tools with those of CofC.

Table 1: Capability comparison of representative catalogue-analysis tools. ✓: native support; (✓): partial support or requires user scripting; –: not supported.

Tool	Multi-format ingest	Duplicate merge	Quality scoring	GR/M_c/ decluster	Web, no install	Provenance/ FAIR export
ZMAP (Wiemer, 2001)	(✓)	–	–	✓	–	–
ObsPy (Beyreuther et al., 2010)	✓	(✓)	(✓)	(✓)	–	(✓)
OpenQuake engine (Pagani et al., 2014)	(✓)	–	–	(✓)	–	–
OQ-MBTK (GEM Foundation, 2024)	✓	✓	(✓)	✓	–	(✓)
SeisComP	(✓)	✓	–	–	–	(✓)
Agency FDSN services	–	–	–	–	✓	(✓)
CofC (this work)	✓	✓	✓	✓	✓	✓

At the institutional level, major agencies, GeoNet (Petersen et al., 2011), the USGS Comprehensive Catalogue (ComCat; U.S. Geological Survey, Earthquake Hazards Program, 2017), the Swiss Seismological Service, the ISC (Storchak et al., 2013), and the EMSC, each publish their own catalogues with FDSN-compliant web services, and the EPOS Seismology Thematic Core Service coordinates access across European networks. ComCat additionally provides a web search interface and the `libcomcat` client for programmatic multi-source retrieval. However, these services expose individual (or centrally pre-merged) agency products rather than providing tools for users to merge, quality-score, and analyse combinations of arbitrary uploaded catalogues from multiple sources. Automated multi-agency event association has been studied in the context of global catalogues (Yeck et al., 2019) and Bayesian magnitude merging across networks (Si et al., 2024), but these methods are not available as general-purpose tools for regional researchers. In New Zealand specifically, Warren-Smith et al. (2025) conducted the first systematic quantitative assessment of GeoNet location quality using azimuthal gap and uncertainty benchmarks, providing the empirical basis for the reliability sub-score in CofC's quality-scoring system (Section 2.6). A high-resolution research catalogue for the Taupo Volcanic Zone (Illsley-Kemp and Mestel, 2025) illustrates the value of specialised regional products that can be ingested alongside the national catalogue, exactly the use case for which CofC is designed.

1.6 FAIR principles and reproducibility

A parallel motivation for CofC is the broader push towards FAIR scientific data: Findable, Accessible, Interoperable, and Reusable (Wilkinson et al., 2016). Wilkinson et al. (2016) codified these principles to address a crisis of reproducibility in data-intensive science, the same crisis that afflicts earthquake catalogue analysis when preprocessing steps are undocumented, format conversions are ad hoc, and processed datasets are not versioned or persistently identified. The FAIR principles have since been adopted as the basis for data infrastructure by the European Plate Observing System (EPOS), which coordinates seismological data services across 24 European nations. For the Aotearoa New Zealand context in particular, FAIR is applied alongside the CARE Principles for Indigenous Data Governance (Carroll et al., 2020), Collective benefit, Authority to control, Responsibility, and Ethics, which recognise Māori data-sovereignty interests in nationally significant datasets; the governance and access-control implications are developed in the companion white paper (Graham et al., 2026).

CofC operationalises the FAIR principles specifically for earthquake catalogue data:

- **Findable.** Every catalogue stored in CofC is assigned a stable internal identifier and indexed, searchable catalogue-level metadata (name, institution, temporal and geographic coverage, magnitude range, event count, licence). Globally unique persistent identifiers are not minted by the platform itself; a released catalogue is instead designed to be deposited with an external agency (e.g. DataCite or Zenodo) to obtain a citable DOI (see Versioning, Section 2).

- **Accessible.** A REST API provides machine-readable access to catalogue metadata and event data in multiple formats (CSV, GeoJSON, QuakeML) with no authentication required for public catalogues.

- **Interoperable.** Data are normalised to a canonical internal schema and re-exportable in standard formats. Magnitude types, event types, and evaluation statuses are stored against controlled vocabularies consistent with QuakeML 1.2 BED (Schorlemmer et al., 2011).

- **Reusable.** Every merged catalogue retains full provenance (source catalogue IDs per event, merge strategy and parameters, quality score). Exports include this metadata, enabling downstream consumers to reconstruct exactly the dataset used in a given study.

The design of CofC was validated through two national community workshops in Aotearoa New Zealand (GSNZ 2023 and GSNZ 2024; Geoscience Society of New Zealand 2023, 2024) that brought together researchers from GNS Science, Victoria University of Wellington, University of Canterbury, and industry. Workshop feedback informed the metadata schema, merge configuration options, quality-grading approach, and access-control model. A companion white paper (Graham et al., 2026) provides the full FAIR framework specification for the Aotearoa New Zealand national earthquake catalogue infrastructure, including governance, licensing, metadata standards, and role-based access control; the present paper focuses on the software implementation and statistical-analysis workflow.

1.7 The Catalogue of Catalogues

CofC addresses these three catalogue-analysis problems, together with the cross-cutting need for transparent quality assessment, in a single, browser-accessible platform. It ingests heterogeneous formats without custom scripting, detects and removes duplicate events using configurable thresholds grounded in the association standards of Storchak et al. (2013) and Engdahl et al. (1998), scores every event on an transparent quality index, and presents GR analysis and declustering in a guided workflow that surfaces the statistical assumptions at each step. Built on Next.js (Vercel Inc., 2024) and MongoDB (MongoDB Inc., 2024), it requires no local installation and is deployable to any cloud provider or on-premises server. The remainder of this paper describes the platform architecture (Section 2), the merging engine (Section 3), the built-in statistical analyses (Section 4), a worked New Zealand example (Section 6), and detailed guidance on avoiding the most consequential statistical biases (Section 7).

1.8 Novelty and contributions

The individual algorithms CofC employs are deliberately standard, maximum-likelihood b-value estimation (Aki, 1965), MAXC completeness estimation (Wiemer and Wyss, 2000), and Gardner-Knopoff (Gardner and Knopoff, 1974) and Reasenberg (Reasenberg, 1985) declustering. The novelty of this work is therefore not a new estimator but the *integration* of the complete multi-catalogue workflow into a single open, web-native, provenance-preserving platform, together with an transparent event-level quality index. Specifically, the contributions are:

1. **An integrated, browser-based workflow** spanning heterogeneous-format ingest, configurable duplicate detection, quality scoring, conflict-resolution merging, and standard seismicity analysis, requiring no local installation or licensed software (Table 1). Catalogue-building libraries such as the GEM toolkit (GEM Foundation, 2024) and operational systems such as SeisComP (GFZ German Research Centre for Geosciences and gempa GmbH, 2008) cover much of the analytical chain, but to our knowledge none delivers it through a browser-native, no-install interface with an integrated per-event quality index.

2. **A transparent, configurable event-level quality index** (equation 1): a five-dimension 0–100 score with letter grades. Its components, location precision, network geometry, solution fit, magnitude quality, and evaluation status, are the standard determinants of solution quality used in agency review and ground-truth location-quality classification (Bondár et al., 2004; Woessner and Wiemer, 2005; Warren-Smith et al., 2025); the contribution is their combination into a single reproducible, user-configurable index that is computed on import and exposed throughout a public web platform, rather than a new quality criterion.

3. **Configurable, automated duplicate detection and conflict resolution** for arbitrary catalogue pairs, with five interchangeable merge strategies and physical-validity gates, generalising the expert-driven association procedures developed for global catalogues (Storchak et al., 2013; Engdahl et al., 1998; Yeck et al., 2019) into a tool regional researchers can apply themselves.

4. **FAIR-by-design provenance**: every merged catalogue records its full lineage (source-catalogue IDs per event, merge strategy and thresholds, quality score) and re-exports it in QuakeML as `comment` elements (Schorlemmer et al., 2011), operationalising the FAIR principles (Wilkinson et al., 2016) for routine catalogue work.

5. **A bias-aware, guided analysis path** that surfaces the underlying statistical assumptions, completeness, magnitude-scale homogeneity, declustering, and quality filtering, at the point of use, consolidating guidance previously dispersed across the CORSSA tutorials (Naylor et al., 2010; Mignan and Woessner, 2012; van Stiphout et al., 2012) into an executable workflow (Section 7).

2 Platform Architecture

CofC presents the catalogue workflow as a single guided pipeline (Figure 1): heterogeneous inputs are ingested and normalised, every event is quality-scored on import, duplicate events are detected and resolved during merging, standard seismicity analyses are run in the browser, and processed catalogues are re-exported in standard formats, with all intermediate state and provenance persisted in the database at every step.

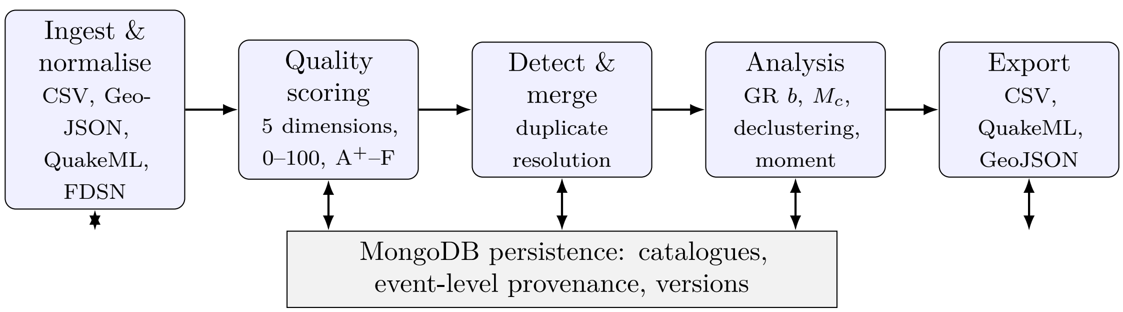

Figure 1: The CofC processing pipeline. Heterogeneous catalogues are ingested and normalised to a canonical schema, quality-scored per event on import (Section 2.6), merged with configurable duplicate detection (Section 3), analysed in the browser (Section 4), and re-exported in standard formats (Section 5). All stages read and write a MongoDB store that retains catalogues, event-level provenance, and version metadata.

2.1 Technology stack

CofC follows a three-tier architecture: a React 18 / Next.js 14 (Vercel Inc., 2024) frontend served via the App Router, a serverless backend layer of Next.js API routes, and a MongoDB 7 (MongoDB Inc., 2024) persistence tier. All tiers communicate through a versioned REST API; no separate application server is required.

- **Frontend:** React 18, TypeScript, Tailwind CSS, and a component library of 40+ shadcn/ui primitives for forms, tables, dialogs, and data-display cards. Recharts provides interactive time-series panels; Leaflet (Agafonkin, 2024) provides the map layer.

- **Backend:** Next.js API routes deployed as serverless functions, enabling horizontal scaling without managing long-running processes. MongoDB 7 with a geospatial index on event coordinates supports efficient bounding-box and radius queries.
- **Parsers:** Custom TypeScript parsers for CSV/TXT, GeoJSON (Butler et al., 2016), and QuakeML 1.2 BED (Schorlemmer et al., 2011). FDSN web-service integration streams live GeoNet data directly into a new catalogue.

The application is containerised with Docker (Compose recipes for development and production) and is designed to deploy to any cloud provider or on-premises server.

2.2 Data model

Each *catalogue* is a named container with metadata (name, description, source agency, status). Each *event* record stores the mandatory seismological fields (origin time, latitude, longitude, depth, magnitude) plus up to 40 optional fields drawn from QuakeML: uncertainty estimates, azimuthal gap, station and phase counts, magnitude type, evaluation mode and status, agency ID, focal mechanisms, phase picks, arrivals, and amplitude measurements. All QuakeML complex sub-objects are stored as JSON strings so that no information is discarded on import.

Magnitude types, evaluation modes, evaluation statuses, and event types are stored lower-cased and validated against controlled vocabularies consistent with QuakeML 1.2 BED, ensuring cross-catalogue comparability.

2.3 Versioning and provenance

CofC adopts a three-tier versioning convention (MAJOR.MINOR.PATCH) following semantic-versioning conventions and DataCite data-versioning best practice (DataCite Metadata Working Group, 2021), under which version labels are incremented according to the following policy:

- **Major** increments when event parameters of existing events change (e.g., full reprocessing with a new velocity model or magnitude calibration).
- **Minor** increments when new events or fields are added without changing existing solutions.
- **Patch** increments for metadata corrections or documentation updates that do not affect event parameters.

Each major and minor release is intended to be deposited with an external archive (e.g. Zenodo) to obtain a persistent DOI, enabling citation of the exact data state used in a given study; DOI minting is not performed by the platform itself. Every merged catalogue carries event-level provenance: source catalogue IDs, the merge strategy and threshold parameters applied, and the quality score at the time of merging. This provenance is recorded as QuakeML `comment` elements (Schorlemmer et al., 2011) on the merged events (and in the catalogue metadata) and supports the reproducibility goals of CSEP-based forecast evaluations (Schorlemmer et al., 2007).

2.4 Visualisation

CofC provides four interactive visualisation views accessible from every catalogue page.

Interactive map. The map view is built on Leaflet (Agafonkin, 2024) and renders events as circles whose radius scales with magnitude and whose colour can encode focal depth, quality grade, or source catalogue (the last is used in Figure 3). For catalogues exceeding 2 000 events, Leaflet's marker-clustering plugin automatically aggregates nearby events into spiderfied cluster markers to maintain browser responsiveness. Two additional overlays are available on demand: (i) *uncertainty ellipses*, which draw the horizontal location error as a scaled ellipse derived from the latitude and longitude uncertainty fields, giving an immediate visual indication of poorly-constrained events; and (ii) *focal-mechanism beach balls*, rendered as SVG quadrant diagrams from the focal-mechanism fields parsed from QuakeML, allowing rapid inspection of faulting style across a catalogue.

Station coverage. The station-coverage panel visualises per-event azimuthal gap as a colour ramp overlaid on the map, complementing the aggregate gap histogram (Figure 4). Events with gap $> 180°$ are highlighted, enabling spatial identification of regions where network geometry is insufficient for reliable location estimates.

Frequency-magnitude distribution. The FMD panel plots the cumulative and non-cumulative event counts by magnitude bin, overlays the fitted GR line and M_c estimate, and updates dynamically when the user adjusts the magnitude cut-off.

Time-series panels. Recharts-based panels display: cumulative event count over time, magnitude-vs.-time scatter, and cumulative seismic moment release. A configurable binning interval (day, week, month) converts the magnitude-time series to a seismicity-rate time series, making step changes in network completeness immediately visible as rate anomalies.

2.5 Supported input formats

Table 2 summarises the five supported formats.

Table 2: Input formats supported by CofC.

Format	Extension(s)	Notes
CSV / TXT	.csv, .txt	Flexible column mapping via template system; supports ∼40 field names per column.
GeoJSON	.geojson, .json	RFC 7946 (Butler et al., 2016) Feature or FeatureCollection; properties auto-mapped.
QuakeML 1.2 BED	.xml, .quakeml	Full preferred-origin / preferred-magnitude extraction; all nested arrays preserved.
GeoNet live import	(FDSN WS)	Configurable date range; streams directly into a new catalogue.

2.6 Quality scoring

Upon import every event receives a quality score $Q \in [0, 100]$ computed as a weighted sum over five dimensions of solution quality:

$$Q = w_{\text{loc}} Q_{\text{loc}} + w_{\text{net}} Q_{\text{net}} + w_{\text{sol}} Q_{\text{sol}} + w_{\text{mag}} Q_{\text{mag}} + w_{\text{eval}} Q_{\text{eval}}, \tag{1}$$

with default weights $w_{\text{loc}} = 0.35$, $w_{\text{net}} = 0.25$, $w_{\text{sol}} = 0.15$, $w_{\text{mag}} = 0.15$, $w_{\text{eval}} = 0.10$. The weights are largest for the dimensions that most directly govern whether a solution is usable in quantitative analysis: location precision and network geometry are the primary determinants of hypocentral accuracy (Bondár et al., 2004; Warren-Smith et al., 2025); the arrival-time fit and magnitude quality are secondary; and the analyst-evaluation status, informative but partly redundant with the instrumental metrics, carries the least weight. Each dimension applies an explicit penalty when its underlying fields are absent, so events with sparse metadata are scored conservatively. The weights are intentionally round-valued and are exposed as configurable parameters of the scoring module (defaults above), so that communities with different priorities can adjust them without modifying the analysis code. Field completeness, physical-consistency checks, and magnitude-scale homogeneity are handled separately by the cross-field validation suite (Section 2.7). The five dimension scores $Q_d \in [0, 100]$ are:

$Q_{\mathbf{loc}}$: **Location (35%)** Solution precision from the reported horizontal, depth, and time uncertainties, weighted most heavily on horizontal uncertainty. Missing uncertainty fields incur a partial penalty, so unconstrained solutions cannot score full marks.

$Q_{\mathbf{net}}$: **Network geometry (25%)** Azimuthal gap, used-station count, and used-phase count. Azimuthal gap is among the principal predictors of horizontal location error (Bondár et al., 2004; Warren-Smith et al., 2025): a gap $\leq 90°$ with ≥ 20 stations earns full points, whereas a gap $\geq 180°$ or very few stations is heavily penalised. GeoNet location quality has been benchmarked against exactly these criteria (Warren-Smith et al., 2025).

$Q_{\mathbf{sol}}$: **Solution fit (15%)** The RMS arrival-time residual (standard error) of the location: < 0.3 s earns full points, with progressively larger penalties above 0.5 and 1.0 s.

$Q_{\mathbf{mag}}$: **Magnitude (15%)** Magnitude uncertainty and the number of stations contributing to the magnitude; small uncertainties and ≥ 10 amplitude measurements earn full points.

$Q_{\mathbf{eval}}$: **Evaluation (10%)** Evaluation mode and status: manually reviewed or final solutions score highest; automatic or preliminary solutions are penalised, reflecting the additional confidence conferred by analyst review.

Figure 5 summarises the five dimensions, their weights, and the grade mapping; the score Q maps to the letter grades in Table 3. Grades are displayed per event on the map and table views, and aggregated at catalogue level.

2.7 Cross-field validation

CofC applies a suite of physical-consistency checks at upload time and presents a summary before the user commits to saving:

Table 3: Quality grade thresholds. The A$^+$–F scale provides an at-a-glance, ordinal event-level quality indicator derived from the continuous score Q of equation (1).

Grade	**Score**	**Recommended use**
A$^+$	95–100	Publication-ready; suitable for PSHA and all quantitative analyses.
A	85–94	Good; suitable for most research and engineering applications.
B$^+$	75–84	Acceptable; document limitations in methods.
B	65–74	Fair; notable gaps in uncertainty or provenance.
C	45–64	Poor; expert review recommended before quantitative use.
D	35–44	Minimal utility for quantitative analysis.
F	<35	Not recommended for quantitative analysis.

- Magnitude-depth plausibility (e.g., $M > 8$ at depth $< 5\,\text{km}$ is extremely rare).
- Depth uncertainty exceeding twice the depth value.
- Magnitude uncertainty exceeding the magnitude for $|M| \geq 1$ (physically implausible).
- Azimuthal gap $> 180°$ (likely station-clustering artefact).
- RMS travel-time residual $< 0.001\,\text{s}$ (suspiciously small, may indicate synthetic or processed data).
- Location-uncertainty anisotropy ratio > 5 (strongly elongated error ellipse suggesting poor azimuthal coverage or geometry).

Each check reports a severity level (*error*, *warning*, *info*) and a remediation suggestion. Errors block ingestion of the affected event; warnings are displayed for review; info flags are informational.

3 Catalogue Merging

3.1 Duplicate detection

Duplicate detection uses an adaptive time-and-distance matching window. Two events from different source catalogues are considered duplicate candidates when both

$$|\Delta t| \leq \tau, \tag{2}$$

$$d \leq \delta, \tag{3}$$

where d is the great-circle distance computed from coordinates and the baseline windows $\tau \leq 60\,\mathrm{s}$ and $\delta \leq 50\,\mathrm{km}$ are widened automatically for larger and deeper events (up to roughly $3\times$ in time, $4\times$ in distance, and $1.5\times$ for depth). Magnitude is not part of the matching test; magnitude consistency is instead enforced as the per-group range gate described below. Near-antimeridian pairs use unit-vector averaging to avoid sign-wrapping errors in the Pacific region.

The baseline windows are user-configurable in the merge UI (with strict, moderate, and loose presets) and follow international practice for global earthquake association (Storchak et al., 2013; Engdahl et al., 1998).

Candidate groups that violate magnitude-scaled physical-validity gates are flagged as merge conflicts and excluded from automated resolution. A group is rejected if its magnitude range exceeds 0.5 ($M < 4$), 0.8 (M 4–5.5), 1.2 (M 5.5–7), or 1.5 ($M \geq 7$); if its spatial spread exceeds 100 km ($M < 5$), 150 km (M 5–6), or 200 km ($M \geq 6$); or if more than 15 events fall in a single group. A network-mismatch guard further prevents merging two distinct events reported by the same network.

3.2 Merge strategies

Once duplicate groups are identified, a conflict-resolution strategy is applied. CofC offers five strategies (Table 4).

The Quality-based strategy is recommended because it selects the best-constrained solution without subjective ranking of agencies, and it preserves all provenance information (source catalogue IDs for every event).

4 Statistical Analyses

4.1 Gutenberg-Richter b-value

The Gutenberg-Richter (GR) frequency-magnitude relation (Gutenberg and Richter, 1944) is

$$\log_{10} N(M) = a - bM, \tag{4}$$

where $N(M)$ is the cumulative number of earthquakes with magnitude $\geq M$, and $b \approx 1$ for tectonic seismicity.

CofC estimates b using the maximum-likelihood estimator of Aki (1965):

$$\hat{b} = \frac{\log_{10} e}{\bar{M} - M_{\min} + \Delta M/2}, \tag{5}$$

where $\bar{M}$ is the sample mean magnitude, $M_{\min}$ is the lower magnitude cut-off (set to the estimated M_c; see Section 4.2), and $\Delta M = 0.1$ is the binning interval (Bender, 1983). The standard error on $\hat{b}$ is (Aki, 1965)

$$\sigma_b = \frac{\hat{b}}{\sqrt{N}}. \tag{6}$$

Equation (6) is the asymptotic counting-statistics error and *understates* the true uncertainty: it ignores magnitude-binning and finite-sample effects captured by the estimator of Shi and

Table 4: Merge strategies available in CofC.

Strategy	**Description**
Quality-based (recommended)	Scores each candidate on equation (1) and retains the highest-scoring event (evaluation status contributes to the score rather than acting as a separate tie-break).
Priority-based	A designated primary catalogue always wins; duplicates from other catalogues are discarded. Useful when one agency is the recognised authority.
Newest data	The event with the most recent origin time is kept. Suitable when later analyses supersede earlier determinations.
Most complete	The event with the greatest number of non-null fields is kept, maximising metadata density in the merged product.
Average values	Latitude and longitude are combined by inverse-variance (uncertainty-weighted) averaging, so lower-uncertainty solutions contribute more; magnitude is taken from the highest-priority ISC/IASPEI type (Mw $>$ Ms $>$ mb $>$ ML $>$ Md) rather than averaged, to avoid scale saturation; depth is taken from the lowest-uncertainty solution and origin time is the earliest in the group. Best treated as exploratory, for closely consistent solutions.

Bolt (1982), and, more importantly, it excludes the systematic sensitivity of $\hat{b}$ to the choice of M_c, to magnitude-scale heterogeneity, and to declustering, which together typically dominate (Section 7). We therefore treat the formal σ_b of Equation (6) as a lower bound and caution that realistic uncertainties are larger. CofC computes $\sigma_b = \hat{b}/\sqrt{N}$ and returns it with each Gutenberg-Richter estimate, displayed alongside the GR plot (cumulative and non-cumulative) and the fitted line.

4.2 Magnitude of completeness

The magnitude of completeness M_c is estimated using the Maximum Curvature (MAXC) method (Wiemer and Wyss, 2000):

$$M_c^{\mathrm{MAXC}} = \arg\max_M \; N_{\text{non-cum}}(M), \tag{7}$$

i.e. the magnitude bin containing the maximum number of events in the non-cumulative frequency-magnitude distribution. MAXC is fast and robust but tends to underestimate M_c by ~ 0.1–0.2 magnitude units for typical networks (Woessner and Wiemer, 2005). CofC applies a $+0.2$ correction by default, which users can adjust.

The formal uncertainty on M_c^{MAXC} is at least ± 0.1 (one bin width), but Woessner and Wiemer (2005) showed empirically that the true uncertainty is typically 0.2–0.3 magnitude units because the non-cumulative FMD peak is broad and sensitive to sample fluctuations. CofC displays the bin-width uncertainty as a lower bound and recommends visual verification against the FMD plot. For high-precision M_c estimation, the Goodness-of-Fit Test (GFT) or Entire Magnitude Range (EMR) method of Woessner and Wiemer (2005) should be applied as external post-processing.

A commonly adopted minimum for a reliable b-value estimate is $N \approx 200$ events above M_c (cf. Marzocchi and Sandri 2003); below this the formal $\sigma_b = \hat{b}/\sqrt{N}$ already exceeds 0.07, comparable to the biases discussed in Section 7, and the confidence interval on b becomes too wide for quantitative hazard use. CofC enforces hard floors below which an estimate is withheld rather than reported, requiring at least 10 events (and three populated magnitude bins) for a Gutenberg-Richter fit and at least 50 events for M_c estimation; the $N \approx 200$ figure is a guideline the analyst should apply when interpreting results.

4.3 Declustering

Aftershock sequences violate the Poissonian independence assumption underlying GR analysis. CofC implements the Gardner-Knopoff window method (Gardner and Knopoff, 1974) with the commonly tabulated magnitude-dependent windows (as compiled by van Stiphout et al. 2012). For each event the space-time interaction zone is:

$$L(M) = 10^{0.1238M+0.983}\ \mathrm{km}, \tag{8}$$

$$T(M) = \begin{cases} 10^{0.5409M-0.547}\ \mathrm{days}, & M \geq 6.5, \\ 10^{0.032M+2.7389}\ \mathrm{days}, & M < 6.5, \end{cases} \tag{9}$$

the two-branch time window $T(M)$ following the larger-magnitude relation for $M \geq 6.5$ and the small-magnitude relation otherwise. Events falling within $L(M)$ and $T(M)$ of a larger mainshock are flagged as aftershocks. The Reasenberg (1985) algorithm, which uses an interaction zone based on the Omori–Utsu law (Omori, 1894; Utsu, 1961) and adapts to the local seismicity rate, is available as an alternative.

The declustered catalogue is exported as a separate catalogue within the platform, with each event tagged as `mainshock` or `aftershock` in the metadata. Window-based methods such as Gardner-Knopoff do not prospectively identify foreshocks; events preceding a larger mainshock within the interaction zone may be retrospectively reclassified as `foreshock` by a post-processing pass, but this reclassification is not applied by default because it is history-dependent and therefore not reproducible in real-time analysis (van Stiphout et al., 2012).

4.4 Cumulative seismic moment

Cumulative seismic moment is computed as

$$M_0 = 10^{1.5M_w+9.1}\ \mathrm{N\,m} \tag{10}$$

using the Hanks-Kanamori relation (Hanks and Kanamori, 1979; Kanamori, 1977) in SI units (equivalently $M_w = \frac{2}{3}\log_{10} M_0 - 6.07$ for M_0 in N m). For events recorded with non-M_w magnitude types, CofC's conversion routines follow the empirical global relations of Scordilis (2006); automatic homogenisation of mixed magnitude types within the analysis pipeline remains future work (Section 7). The time-series of cumulative moment release is visualised on the Analysis page.

4.5 Temporal pattern analysis

Beyond the aggregate GR analysis, CofC provides two temporal diagnostics aimed at detecting non-stationarity in the catalogue.

Seismicity rate time series. Events above the estimated M_c are binned at a user-selected interval (day, week, or month) and plotted as a rate time series. Sudden rate changes, which often correspond to network upgrades, station outages, or changes in processing procedures, are immediately visible and can be used to identify breakpoints for time-varying M_c analysis (Mignan and Woessner, 2012).

Interevent-time distribution. The distribution of times between successive events above M_c is plotted against the exponential distribution expected for a Poissonian process. Systematic departures, most commonly the excess of short interevent times produced by aftershock clusters, provide a visual diagnostic of whether the catalogue has been adequately declustered before analyses that assume Poissonian seismicity.

5 Export Formats

Processed or merged catalogues can be exported in:

- **CSV**: flat table with all fields, suitable for ZMAP (Wiemer, 2001), ObsPy (Beyreuther et al., 2010), and R/Python workflows.
- **QuakeML 1.2 BED** (Schorlemmer et al., 2011): re-serialisation of the parsed origin, magnitude, and extended fields, with merge provenance recorded as QuakeML `comment` elements.
- **GeoJSON** (Butler et al., 2016): suitable for GIS software and web mapping.

6 Worked Example: New Zealand Seismicity

To illustrate the CofC workflow and quantify the statistical biases described in Section 7, we construct a *synthetic* catalogue pair representative of two agencies monitoring New Zealand seismicity. All event parameters (location, depth, magnitude, azimuthal gap) are drawn from statistical distributions calibrated to GeoNet observations (Petersen et al., 2011; Warren-Smith et al., 2025) but do not represent any specific real dataset. This controlled synthetic setting allows us to demonstrate bias magnitudes in a transparent, reproducible way; we note where results are expected to generalise and where real-data validation is needed. The two synthetic catalogues (GeoNet-like and Agency B) span January 2020–December 2024 and contain

$\sim$120 000 and $\sim$98 000 events respectively, matching the scale of the real GeoNet catalogue over that period (Petersen et al., 2011).

6.1 Step 1: Import

Both catalogues are uploaded as QuakeML. CofC reports per-catalogue statistics immediately: event counts, magnitude range, temporal span, and completeness grade distribution. Cross-field validation flags $\sim$2% of Agency B events for unusually large depth uncertainties ($\sigma_z > 2 \times z$), consistent with poorly constrained shallow focal depths near the Hikurangi subduction interface.

6.2 Step 2: Quality assessment

The median quality score is 72 (grade B) for GeoNet and 58 (grade C) for Agency B, reflecting GeoNet's denser station network and manual review procedures. The azimuthal gap distribution (Figure 4) shows a long tail $> 180°$ in Agency B, confirming station-coverage limitations for offshore events.

6.3 Step 3: Merge

Applying the Quality-based strategy with default thresholds (equations 2 and 3) identifies 49 000 duplicate pairs (50% of Agency B events have a GeoNet counterpart); Figure 3(a) illustrates the duplicate-matching geometry, in which the same earthquake reported by both agencies falls within the matching window. Concatenating the $120\,000 + 98\,000 = 218\,000$ raw records and removing one member of each duplicate pair yields a merged catalogue of 169 000 unique events. Provenance is retained (Figure 3b): 71 000 events are from GeoNet only, 49 000 from Agency B only (events outside GeoNet's detection footprint or below its reporting threshold), and 49 000 from the Quality-based resolution of duplicate pairs. Because the overlap is injected, this example provides exact ground truth for the matched pairs and here functions as a consistency check on the merge bookkeeping rather than a benchmark of detection accuracy. Quantifying duplicate-detection precision and recall on genuinely heterogeneous real catalogues, where location scatter is non-Gaussian and clocks and magnitude scales differ, and validating the quality score against an independent benchmark such as Warren-Smith et al. (2025) are left to future work (Section 9).

6.4 Step 4: Quality filtering and statistical analysis

Applying a quality threshold of $Q \geq 70$ (upper grade B and above) to the merged catalogue retains 143 650 events (85% of the 169 000 unique events). The 25 350 removed events are predominantly offshore Agency B sources with azimuthal gap $> 180°$ and $\sigma_{\mathrm{loc}} > 10$ km, exactly the population visible in the tail of Figure 4. Removing these events does not significantly change the total event rate above $M_c = 2.0$ for onshore New Zealand, but it eliminates a spatially biased subset that would otherwise distort spatial b-value maps along the Hikurangi margin.

Gutenberg-Richter b-value estimation on the quality-filtered catalogue (Section 4), restricted to the 48 600 of the 143 650 retained events at or above $M_c = 2.0$ (estimated by Maximum Curvature with the standard +0.2 correction), gives:

$$\hat{b} = 1.02 \pm 0.005 \quad (M_c = 2.0,\ N = 48\,600)$$

which by construction matches the value to which the synthetic quality-filtered catalogue was calibrated (chosen near the realistic New Zealand value $b \approx 1.0$; cf. Cattania et al. 2018). This is a check that the estimator and M_c handling behave correctly on a controlled input, not an empirical measurement of New Zealand seismicity. Full provenance is preserved throughout: each event in the export carries its source catalogue ID(s), the quality score, and the merge strategy applied.

As an additional preprocessing step, applying Gardner-Knopoff declustering (Section 4.3) reduces the catalogue by a further 23%, removing the clustered aftershock population and lowering $\hat{b}$ slightly to 0.94 ± 0.005 (Figure 6), illustrating the sensitivity of b-value estimates to preprocessing choices (Section 7). Figure 2 summarises the event-count flow through these steps.

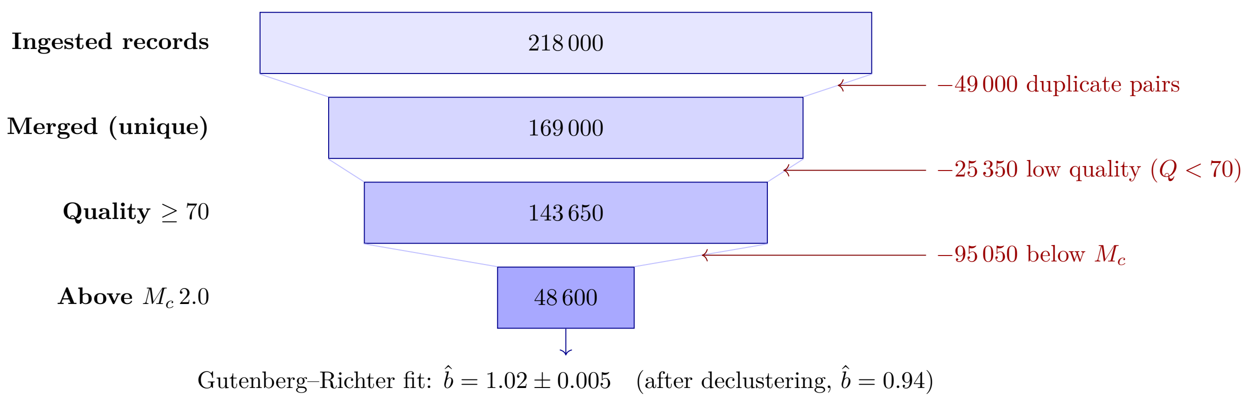

Figure 2: Quantitative event-count flow through the CofC pipeline for the synthetic worked example (Section 6). Bar widths are proportional to the number of events; each stage removes a characterised subset (red) before the Gutenberg–Richter b-value is estimated on the surviving population. All counts are properties of the controlled synthetic inputs, not real New Zealand seismicity.

7 Guidance on Avoiding Statistical Biases

This section distils the most consequential biases that arise when working with merged multi-agency catalogues. They are not unique to CofC; they affect any merged catalogue analysis.

7.1 Magnitude completeness bias

Problem. The GR relation is only linear above M_c. Fitting below M_c underestimates b because the incomplete sample is deficient at small magnitudes.

Guidance.

1. Always estimate M_c before fitting b. Use the CofC MAXC estimate as a starting point but verify by eye on the FMD plot.

2. Set the lower magnitude cut-off at $M_c + 0.1$ to further protect against partial completeness in the transition zone (Woessner and Wiemer, 2005).

3. After merging catalogues with different M_c values, use the *higher* of the two as the common cut-off to guarantee completeness across the merged set.

7.2 Magnitude heterogeneity bias

Problem. Mixing magnitude scales (e.g., ML from one agency, Mw from another) artificially broadens the FMD and can shift the apparent b-value. Local magnitude ML saturates above $M \approx 6.5$; body-wave magnitude mb saturates above $M \approx 6.0$ (Bormann and Saul, 2009).

Guidance.

1. Inspect the magnitude-type distribution in the Analysis page before fitting.

2. Convert to a single preferred scale (ideally Mw) using empirical regressions (Das et al., 2011; Hanks and Kanamori, 1979), preferably orthogonal rather than ordinary least-squares fits, to avoid conversion bias (Castellaro et al., 2006; Lolli and Gasperini, 2012), where possible.

3. If conversion is not possible, split the catalogue by magnitude type and fit each subset separately.

7.3 Network coverage and azimuthal gap bias

Problem. Events in sparse network regions have systematically larger location and depth uncertainties. If these low-quality events are preferentially at one magnitude or depth range they bias statistical distributions.

Guidance.

1. Filter on quality score (e.g., $Q \geq 50$, grade C or better) or azimuthal gap ($< 180°$) before statistical analysis.

2. Check the spatial distribution of excluded events; geographic concentration may indicate a bias rather than a data problem.

3. Use the CofC map colour-coded by quality grade to identify network-coverage artefacts.

7.4 Aftershock clustering and GR analysis

Problem. GR analysis assumes independent, Poissonian arrivals. Aftershock sequences cluster in space, time, and magnitude, violating this assumption and shifting the estimated b, in the synthetic example of Section 6 the quality-filtered catalogue gives $\hat{b} = 1.02$ before declustering versus 0.94 after, an upward bias in the un-declustered value.

Guidance.

1. Always decluster before GR fitting. The Gardner-Knopoff method (Gardner and Knopoff, 1974) implemented in CofC is the standard choice for tectonic seismicity; Reasenberg (1985) is preferable for swarm-dominated sequences.

2. Export both the original and declustered catalogues and compare b-values to quantify the bias.

3. Report the declustering algorithm and parameters in publications so results are reproducible.

7.5 Duplicate double-counting

Problem. Merging without duplicate removal inflates $N(M)$ for all magnitudes above the lower-agency M_c, systematically lowering the estimated a-value and distorting rate estimates.

Guidance.

1. Always merge catalogues before statistical analysis; never concatenate raw files.

2. Inspect the merge summary (duplicate count, strategy applied) and satisfy yourself that the matching thresholds are physically appropriate for your region and magnitude range.

3. For teleseismic analysis, increase the distance threshold (e.g., to 100 km) to account for larger location scatter at global distances.

7.6 Temporal non-stationarity

Problem. Network upgrades, station outages, or changes in analysis procedures alter M_c over time. A single b fitted over a long catalogue period conflates eras with different completeness.

Guidance.

1. Use the CofC time-series view to identify sudden jumps in event rate or minimum magnitude, likely indicators of network changes.

2. Split the catalogue at breakpoints and estimate b for each period separately.

3. The Mignan and Woessner (2012) review provides detailed methods for time-varying M_c analysis.

8 Software Availability

CofC is released under the MIT licence (a `LICENSE` file and a `CITATION.cff` are included in the repository). The source code, issue tracker, and documentation are available at:

https://github.com/KennyGraham1/catalogofcatalogs

Full documentation (user guide, developer guide, API reference, deployment guide) is hosted at:

https://catalogofcatalogs.readthedocs.io/en/latest/

Docker Compose recipes for single-command deployment, and a continuous-integration pipeline (linting, type-checking, and the Jest test suite on Node 18 and 20), are included in the repository. For citation of a fixed software state, a tagged release is archived with a Zenodo DOI (assigned at submission); the present paper describes release `v0.1.0`.

System requirements

- Node.js $\geq$ 18.x
- MongoDB $\geq$ 6.x (local or Atlas cloud)
- Modern browser (Chrome, Firefox, Edge, Safari)

Dependencies

Key third-party dependencies (all MIT or compatible licences): Next.js (Vercel Inc., 2024), MongoDB Node.js driver, Leaflet (Agafonkin, 2024), Recharts, Tailwind CSS, shadcn/ui, xml2js (QuakeML parser), uuid.

9 Discussion and Conclusions

9.1 Contribution in the context of existing approaches

CofC addresses a gap that is simultaneously methodological and infrastructural. The methodological gap is well documented in the CORSSA literature (Naylor et al., 2010; Mignan and Woessner, 2012; van Stiphout et al., 2012): the biases described in Section 7 are known, quantifiable, and avoidable, yet they continue to appear in published seismicity studies because the corrective steps, enforcing M_c, homogenising magnitude scales, declustering, and quality filtering, are each individually inconvenient to apply and are rarely performed together in a reproducible workflow. The infrastructural gap is that no production-ready tool integrates all of them: ZMAP (Wiemer, 2001) handles GR analysis and spatial b-value mapping but not merging or quality scoring; ObsPy (Beyreuther et al., 2010) handles I/O and QuakeML parsing but leaves the statistical workflow entirely to the user; the ISC catalogue (Storchak et al., 2013) provides expert-reviewed global data but is static (updated annually) and not directly linkable to regional agency data for local analyses. CofC closes this loop, from multi-source ingest through duplicate removal, quality scoring, declustering, and GR fitting, within a persistent web interface that preserves provenance at every step.

9.2 Quality scoring and event selection

The quality-scoring system (equation 1) is designed to be interpretable: each sub-score corresponds to a well-understood determinant of solution quality, grounded in the seismological literature. Azimuthal gap is among the principal network-geometry predictors of horizontal location error (Bondár et al., 2004); used-station and used-phase counts further constrain the solution; location and magnitude uncertainties are direct quantitative measures of solution precision. By making these criteria explicit and combining them into a single 0–100 index, CofC provides a reproducible basis for quality-based event selection that does not depend on subjective judgement of agency provenance.

An alternative quality-filtering approach common in the literature is to apply hard thresholds on individual parameters (e.g., retain only events with gap $< 180°$ and $\sigma_M < 0.3$). Such thresholds are easier to report in a methods section but are less flexible: a threshold approach discards events entirely, whereas the quality score allows users to weight the trade-off between sample size and data quality continuously. CofC exposes both approaches: the quality score for continuous ranking and per-field uncertainty filters in the advanced filter API.

9.3 Towards reproducible seismicity analysis

A persistent challenge in observational seismology is that two researchers applying the same analysis method to nominally the same dataset can obtain different results, because their preprocessing choices, format conversion, duplicate removal, M_c cut-off, declustering algorithm and parameters, quality filters, differ in ways that are rarely fully reported (Schorlemmer et al., 2007). CofC attacks this problem at the infrastructure level: every catalogue stored in the platform carries a complete provenance record (source catalogue IDs per event, merge strategy and parameters, quality score, declustering algorithm and tagging). Every export includes this metadata, and the QuakeML export format (Schorlemmer et al., 2011) provides a standard container for sharing it. The platform thus encourages a workflow in which the preprocessing choices are not buried in bespoke scripts but are captured as persistent, shareable catalogue objects.

9.4 Limitations and future work

M_c estimation. The MAXC method (Wiemer and Wyss, 2000) implemented in CofC is fast and requires no assumptions about the GR model, but Woessner and Wiemer (2005) showed that it systematically underestimates M_c by ~ 0.1–0.2 magnitude units in typical network conditions. The +0.2 correction applied by default mitigates this but does not eliminate it. The Goodness-of-Fit Test (GFT) and Entire-Magnitude Range (EMR) methods of Woessner and Wiemer (2005) are more accurate, particularly for catalogues with gradual network growth, and will be added in a future release. For time-varying M_c, the bootstrap approach described by Mignan and Woessner (2012) is planned.

Spatial analysis. CofC currently provides only global (single value) b-value estimates. Spatial mapping of b (Wiemer and Wyss, 2000; Schorlemmer et al., 2005), of M_c (Schorlemmer and Woessner, 2008), and of seismicity rates on a grid or along a fault structure requires server-side spatial indexing and is targeted for a future release. Similarly, ETAS parameter fitting (Ogata, 1988) and forecast evaluation using the CSEP framework and its consistency tests (Schorlemmer et al., 2007; Zechar and Jordan, 2010) are not yet implemented.

Scale. Catalogues larger than $\sim 10^6$ events require server-side pre-aggregation (e.g., hexagonal binning for the map view, streaming computation for GR fits) for smooth browser rendering. MongoDB's aggregation pipeline and server-side sampling, already used for the API pagination layer, will be extended to support these workflows.

Magnitude homogenisation. CofC displays the magnitude type of each event and warns when a catalogue contains mixed scales, but it does not yet apply empirical inter-scale conversion automatically. Given the magnitude bias this can introduce (Marzocchi and Sandri, 2003; Tinti and Mulargia, 1987), automated conversion to a unified scale (ideally M_w) using orthogonal regressions such as those of Das et al. (2011) and Lolli and Gasperini (2012) is a high-priority addition.

9.5 Conclusions

We have presented CofC, a browser-based platform for earthquake catalogue management, merging, quality scoring, and seismological analysis. The core contribution is infrastructural: a single persistent interface that integrates format ingestion, duplicate detection, transparent quality scoring, and provenance-preserving export, the steps that are individually inconvenient yet collectively determine the validity of downstream statistical analysis. Using a fully synthetic two-catalogue New Zealand example, we illustrated the end-to-end workflow and verified that it is internally consistent and reproducible: an injected 50% duplicate overlap is reproduced, quality scoring isolates an offshore high-azimuthal-gap population that would otherwise bias spatial seismicity analysis, and Gutenberg-Richter analysis of the quality-filtered catalogue returns the input $\hat{b} = 1.02$ with full provenance retained. These figures characterise the controlled synthetic inputs and the software's behaviour on them, not real New Zealand seismicity; validation on heterogeneous real catalogues, duplicate-detection precision and recall, and quality-score calibration against independent benchmarks, is the priority next step. By making preprocessing choices explicit and exportable, CofC supports the shift towards reproducible, openly documented seismological practice advocated by the CORSSA community (Naylor et al., 2010) and the FAIR data principles (Wilkinson et al., 2016).

The platform is actively developed; community contributions, feature requests, and bug reports are welcomed via the GitHub issue tracker.

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Data and Resources

The worked example in Section 6 uses *synthetic* seismicity data generated with parameters calibrated to GeoNet statistics; no real event data are used or released. All quantities quoted in Section 6 and the three data figures (Figures 3–6) are reproduced by the seeded script `paper/figures/generate_figures.py` (Python; fixed seed 42) in the source repository, which prints the worked-example summary and writes the figures. The GeoNet FDSN Event Web Service, from which users can obtain the live New Zealand catalogue, is available at https://www.geonet.org.nz/. The ISC-GEM catalogue (Storchak et al., 2013) is available at https://www.isc.ac.uk/iscgem/. All CofC source code and documentation URLs are given in Section 8.

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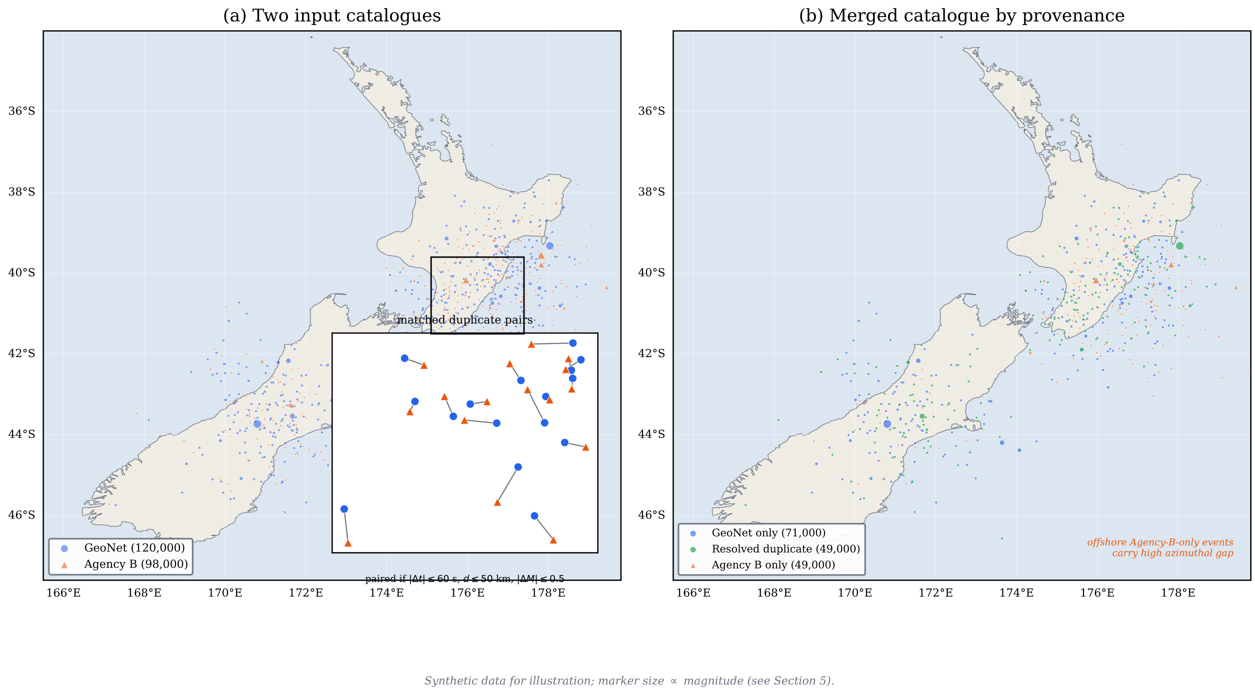

Figure 3: How the merge works, on the synthetic New Zealand example (Section 6). *(a)* The two input catalogues overlaid: the GeoNet-like primary catalogue (blue circles) and the secondary Agency B catalogue (orange triangles). The inset zooms on the boxed region and shows that the same earthquake is reported by both agencies with a small offset; such events are flagged as duplicates when they fall within the time, distance, and magnitude window (equations 2 and 3). *(b)* The merged catalogue coloured by provenance: events seen only by GeoNet (blue), events present in both and resolved to a single solution (green), and events seen only by Agency B (orange). The Agency-B-only events concentrate offshore along the Hikurangi margin, carry high azimuthal gap, and are the population removed by quality filtering (Section 6). Marker size scales with magnitude. Data are synthetic and for illustration only.

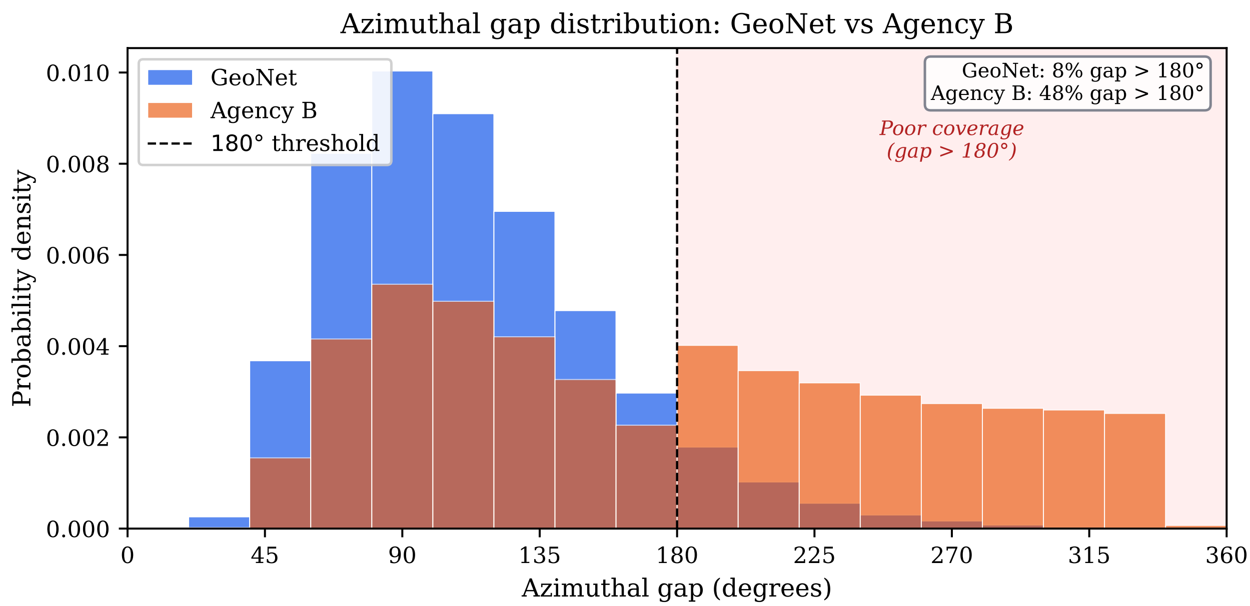

Figure 4: Distribution of azimuthal gap (in degrees) for the GeoNet catalogue (blue) and Agency B (orange) before merging. The long tail $> 180°$ in Agency B reflects limited offshore station coverage. Events with gap $> 180°$ receive a low network-geometry sub-score Q_{net} (Section 2.6) and are typically graded C or below. The shaded region marks the poor-coverage zone. Percentages shown are the fraction of events in each catalogue with gap exceeding 180°.

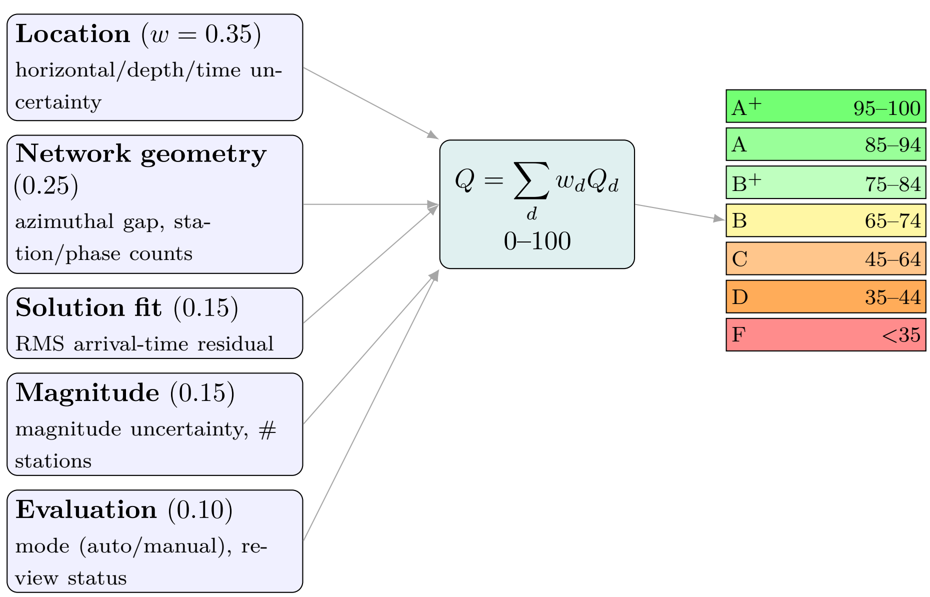

Figure 5: The per-event quality index (Section 2.6). Five dimensions, each computed from the listed event attributes and weighted as shown, are combined into a single score $Q \in [0, 100]$ (equation 1) that maps to an A$^+$–F grade (Table 3; colours match the map view). Missing attributes incur per-dimension penalties, so sparsely-documented events are scored conservatively.

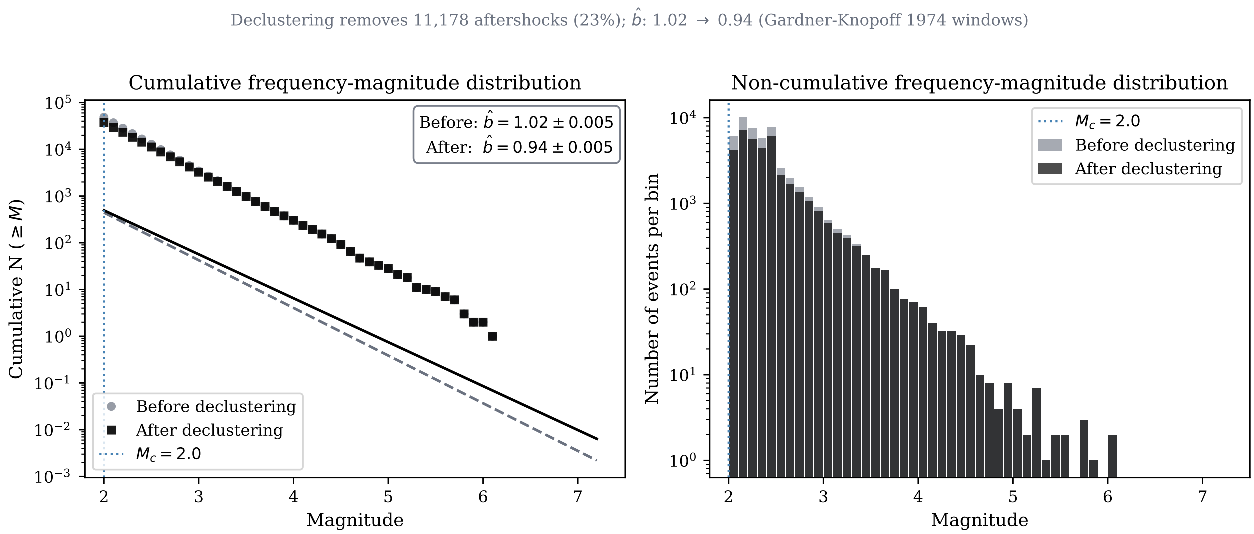

Figure 6: Frequency-magnitude distributions (FMD) for the synthetic New Zealand catalogue before (grey symbols/bars) and after (black) Gardner-Knopoff declustering. *Left:* cumulative FMD with maximum-likelihood GR fits (dashed line: before; solid line: after) above $M_c = 2.0$ (blue dotted line). *Right:* non-cumulative FMD showing the excess of small events contributed by aftershock sequences. Declustering removes these clustered small events, shifting $\hat{b}$ from 1.02 ± 0.005 (before declustering) to 0.94 ± 0.005 (declustered) and removing 23% of events. Data are synthetic and for illustration only (see Section 6).